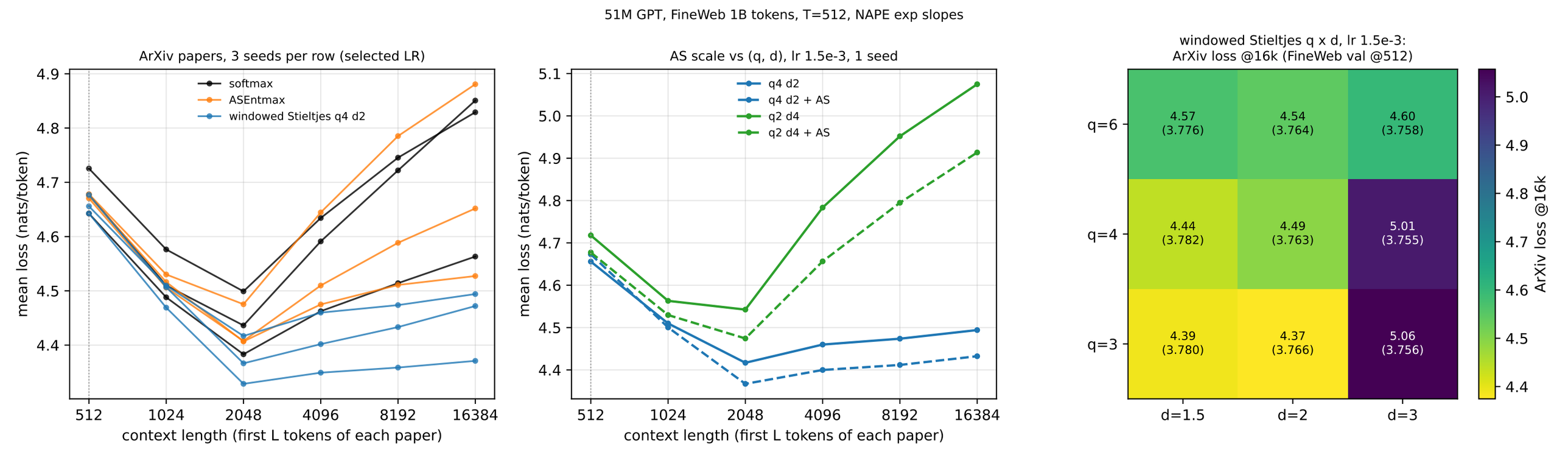
51M GPT, FineWeb 1B tokens, T=512, NAPE exp slopes
ArXiv papers, 3 seeds per row (selected LR)
softmax
ASEntmax
windowed Stieltjes q4 d2
mean loss (nats/token)
context length (first L tokens of each paper)
AS scale vs (q, d), lr 1.5e-3, 1 seed
q4 d2
q4 d2 + AS
q2 d4
q2 d4 + AS
mean loss (nats/token)
context length (first L tokens of each paper)
windowed Stieltjes q x d, lr 1.5e-3:
ArXiv loss @16k (FineWeb val @512)
q=6
q=4
q=3
d=1.5
d=2
d=3
4.57 (3.776)
4.54 (3.764)
4.60 (3.758)
4.44 (3.782)
4.49 (3.763)
5.01 (3.755)
4.39 (3.780)
4.37 (3.766)
5.06 (3.756)
ArXiv loss @16k